

The empirical recurrence for OEIS A235543: recovering a conjecture that is not written down

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Abstract

OEIS A235543 records “Empirical recurrence of order 77”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 252$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 77, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 77 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235543 is “Number of $(n+1) \times (3+1)$ 0..1 arrays with the sum of each 2×2 subblock two median terms lexicographically nondecreasing rowwise and columnwise.”. It has offset 1 and begins

141, 1049, 8711, 61401, 429443, 2704866, 16395328, 93093885, ...

The entry states:

Empirical recurrence of order 77 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:37:07, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 328$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 252$ of the 328, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 252$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 656$ exact terms of the model returns order 77 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	28	c_{21}	731373432604	c_{41}	−2892470592310404	c_{61}	9050280458422
c_2	−302	c_{22}	−3309162219324	c_{42}	1697366883839632	c_{62}	3192216918994
c_3	1226	c_{23}	−830899894248	c_{43}	2694979111165796	c_{63}	−2618034397174
c_4	3426	c_{24}	12455975456382	c_{44}	−2177375903885464	c_{64}	−421558839819
c_5	−52626	c_{25}	−3767137749684	c_{45}	−2087074204220636	c_{65}	585707058846
c_6	129450	c_{26}	−37323379322592	c_{46}	2248422407776572	c_{66}	12941484520
c_7	610518	c_{27}	28226445644400	c_{47}	1308588393688344	c_{67}	−99192700912
c_8	−3904368	c_{28}	90228699322140	c_{48}	−1921005828414933	c_{68}	8844111536
c_9	547530	c_{29}	−108859214758500	c_{49}	−623576843346312	c_{69}	12144019232
c_{10}	48926058	c_{30}	−175287228234708	c_{50}	1373046401401230	c_{70}	−2309774080
c_{11}	−97411422	c_{31}	307359046653816	c_{51}	181127324643062	c_{71}	−976531456
c_{12}	−335286778	c_{32}	265696812054480	c_{52}	−823267655267558	c_{72}	298662656
c_{13}	1384939714	c_{33}	−691470162518432	c_{53}	19227670039030	c_{73}	39346688
c_{14}	922108078	c_{34}	−283955460340504	c_{54}	412858169252522	c_{74}	−21354496
c_{15}	−11488323650	c_{35}	1282333699510028	c_{55}	−65105354329038	c_{75}	348160
c_{16}	6250804033	c_{36}	115896171724932	c_{56}	−171710645629680	c_{76}	675840
c_{17}	65682389734	c_{37}	−1992159037277528	c_{57}	48728062076022	c_{77}	−73728
c_{18}	−95806466040	c_{38}	321619738665088	c_{58}	58315973209902		
c_{19}	−267641333236	c_{39}	2611518947089160	c_{59}	−24224436433374		
c_{20}	674461897208	c_{40}	−990315525411150	c_{60}	−15732394072882		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 77, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $77 + 4$ terms removed returns order 77 as well, so $\deg q_{\min} = 77 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $77 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 77 that this sequence satisfies — and that family turns out to have exactly one member.

References

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